Genesis Directive: Contrastive Latent Plan Coupling for Structure-Sensitive Language Generation

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Abstract

We introduce **Genesis Directive**, a framework for investigating the representation geometry of explicitly coupled latent planning channels in autoregressive language generation. We couple a *diffusion-based latent planner* to an *autoregressive decoder* through a learned projection bridge. By isolating the optimization of semantic compatibility from order-aware structural sensitivity, we uncover a fundamental architectural tension: mechanisms capable of inducing strong structural separation simultaneously destabilize decoder-compatible semantic geometry. Our experiments on WikiText-2 define three distinct representational regimes: (1) *Pure Alignment*, which establishes strong semantic utility ($\Delta_z = +1.61$ nats); (2) *Full Contrastive Optimization*, which induces strong structural sensitivity ($S = +1.37$ nats) but collapses the semantic manifold; and (3) *Orthogonal Dual-Head architectures*, which preserve semantic stability but lack the representational capacity for deep structural separation. We conclude that semantic compatibility and structural discrimination occupy competing optimization regimes in latent-prefix coupling, necessitating deeper architectural unbinding than shallow gating or additive orthogonalization can provide. Code and checkpoints are available at https://github.com/safehavenlabs/genesis-directive.

1 Introduction

Large language models (LLMs) generate text autoregressively, conditioning each token on the preceding context. While powerful, this paradigm offers no principled mechanism for high-level planning prior to generation. Humans, by contrast, tend to form an abstract outline before writing — a semantic skeleton that guides word-by-word realisation.

Diffusion language models such as LLaDA [1] operate in *latent space*, iteratively denoising a full-sequence representation. We ask: are explicit latent planning channels a required inductive bias for coherence under autoregressive constraints? Can an externalized latent structure encode global ordering constraints that an autoregressive decoder actively binds to?

This motivates the **Genesis Directive** framework (Figure 1). Our contributions are:

1. A framework for mapping the representation geometry of structural dependency, pairing a frozen diffusion planner with a LoRA-adapted autoregressive decoder via a learned *projection bridge*.

2. The **Wrong-Structure Penalty** (S), an evaluation metric that isolates structural asymmetry, providing a stable measure of structural dependency regardless of semantic noise floors.

3. An **order-aware self-shuffle contrastive loss** that operationalises global ordering constraints in the latent projection manifold.
4. Experimental mapping of three distinct optimization regimes that empirically prove semantic compatibility and structural discrimination compete for the same latent coordinates.
5. A systematic mixed-precision training protocol for Apple MPS that eliminates fp16 NaN gradients entirely without requiring a decoder backward pass.

2 Background

2.1 Diffusion Language Models

Diffusion LMs define a forward process that corrupts a sequence x_0 with Gaussian or masking noise over T steps, and learn a reverse process to recover the original. LLaDA [1] performs masked diffusion directly over token IDs, producing hidden representations $Z \in \mathbb{R}^{L \times d_p}$ that encode full-sequence semantics without the strict left-to-right inductive bias of autoregressive models.

2.2 Soft Prompting and Prefix Tuning

Prefix tuning [3] prepends learnable continuous vectors to the decoder's embedding sequence, steering generation without fine-tuning base model weights. We extend this idea by replacing the learnable prefix with a *task-conditioned latent plan* produced dynamically by the diffusion planner.

2.3 Contrastive Representation Learning

Contrastive methods encourage representations to be similar for semantically related inputs and dissimilar for unrelated ones. SimCLR [5] and InfoNCE [6] have been widely applied in vision; their application to discrete sequence structure in language models is less explored. Our self-shuffle contrastive loss is a novel adaptation that specifically penalises insensitivity to *token order* within the latent plan.

3 Method

3.1 Architecture Overview

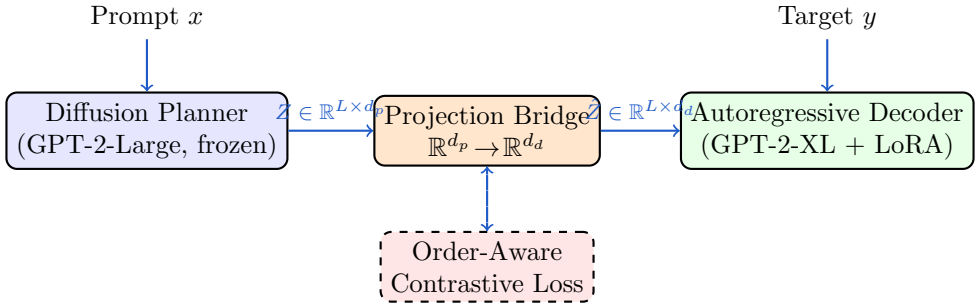

Figure 1: Genesis Directive architecture. The frozen diffusion planner produces a latent plan Z; the projection bridge maps it to the decoder's embedding space; the decoder generates text conditioned on the soft prefix $\hat{Z}$. The contrastive loss trains the bridge in fp32 without a decoder backward pass.

Planner. We use GPT-2-Large [2] as a masked-diffusion surrogate following LLaDA [1]: we iteratively denoise masked tokens using GPT-2 hidden states as the latent trajectory, extracting the final hidden-state sequence as the plan Z. Given a prompt x, the planner produces $Z = f_\theta(x) \in \mathbb{R}^{L\times 1280}$, where L is the sequence length.

Projection Bridge. A learned MLP maps Z from the planner dimension $d_p = 1280$ to the decoder dimension $d_d = 1600$:

$$\hat{Z} = g_\phi(\mathrm{LayerNorm}(Z)), \quad g_\phi : \mathbb{R}^{d_p} \to \mathbb{R}^{d_d}. \tag{1}$$

We evaluate four architectures for g_ϕ (Section 4.5).

Decoder. GPT-2-XL [2] with LoRA adapters [4] ($r = 8$, target `c_attn`) is used as the decoder. The soft prefix is formed by concatenating $\hat{Z}$ with the target text embeddings before the first attention layer:

$$h_0 = \big[\hat{Z};\, \mathrm{Embed}(y)\big]. \tag{2}$$

Labels over the $\hat{Z}$ prefix positions are masked (-100) so the cross-entropy loss is computed only over the target text tokens.

3.2 Order-Aware Contrastive Loss

The key insight of Genesis Directive is that the projection bridge should encode not just *what* tokens appeared in the plan, but *in what order*. We operationalise this with an **order-aware self-shuffle hinge loss**:

$$\mathcal{L}_{\text{contrast}} = \max\Big(0,\ m - \big\|\hat{Z} - \hat{Z}_\pi\big\|_F^2\Big), \tag{3}$$

where $\hat{Z}_\pi = \hat{Z}[:, \pi, :]$ is the projected latent with its token positions permuted by a random permutation π, and m is a margin hyperparameter. The gradient of $\mathcal{L}_{\text{contrast}}$ with respect to ϕ is computed entirely in fp32 without involving the decoder's forward or backward pass, which is critical for numerical stability on mixed-precision hardware.

3.3 Training Protocol

The decoder and planner weights are *frozen* throughout. Only the projection parameters ϕ are updated via AdamW with lr $= 10^{-4}$, gradient clipping at norm 1.0, and gradient accumulation over 8 steps.

Mixed-Precision NaN Prevention. We identified that calling `model.train()` on the full model causes the fp16 SDPA attention in the frozen planner to produce NaN activations on Apple MPS. The following protocol eliminates this entirely:

1. Set `model.eval()` globally.
2. Set `model.projection.train()` only — projection weights are fp32.
3. Compute $\hat{Z}$ and the contrastive loss in fp32.
4. Never backpropagate through the fp16 decoder.
5. Exclude `z_norm` from trainable parameters (the normalization receives gradients from both $\hat{Z}$ and $\hat{Z}_\pi$, doubling the effective gradient magnitude and causing NaN).

4 Experiments

4.1 Setup

Dataset. WikiText-2 [7] (raw, v1). We use the top 10% of the training split filtered to paragraphs with > 30 characters, yielding 1,769 texts. Pairs (x_i, y_i) are formed from consecutive texts. We use 40 pairs for training and 15 for evaluation in initial ablations, scaling to 450 pairs for training and 50 for evaluation in our final representational geometry mapping.

Baselines. Each experiment reports fluency loss under three conditions: **Real**: actual projected plan $\hat{Z}$; **Zero**: $\hat{Z}$ replaced with a zero tensor; **Shuffle**: $\hat{Z}$'s token positions randomly permuted.

Structure Sensitivity Metric. We define the **wrong-structure penalty**:

$$S = \ell_{\text{shuffle}} - \ell_{\text{zero}} \tag{4}$$

$S > 0$ means the decoder penalises wrong structure more than absent structure *regardless of the sign of* $\Delta_z = \ell_{\text{zero}} - \ell_{\text{real}}$. This directly operationalises the headline claim and is safe to use even when zeroing the prefix helps generation. We also report component gaps Δ_z and $\Delta_s = \ell_{\text{shuffle}} - \ell_{\text{real}}$ for interpretability.

Dataset note: We intentionally use a small evaluation set (ranging from 40 to 450 training pairs) to isolate architectural signal rather than maximise generation performance; this is not a benchmark paper.

4.2 Experiment 1: Attention Attribution

To verify that the decoder actually attends to the latent prefix, we extract per-head attention weights and partition mass between the prefix and text positions.

Table 1: Mean attention mass on prefix vs. text (averaged over all heads and layers).

Region	**Attention Mass**
Latent Prefix $\hat{Z}$	0.5349
Text Baseline	0.4651

The decoder devotes **53.5%** of its attention to the latent prefix, confirming that the projection bridge creates an informative soft-prompt that the decoder actively reads.

4.3 Experiment 2: Prefix Ablation (Pre-Training Baseline)

Before any contrastive training, we assess the model's baseline dependence on the latent plan.

Table 2: Pre-training ablation baseline. Positive Δ = ablation hurts generation.

Condition	**Fluency Loss**	**Δ vs. Real**
Real $\hat{Z}$	6.946	—
Zeroed $\hat{Z}$	4.856	-2.090
Shuffled $\hat{Z}$	7.238	$+0.293$

The zero ablation *helps* the decoder (lower loss), suggesting the untrained projection injects noise that interferes with generation. Shuffling the plan already causes a +0.29 nat increase in loss, indicating the decoder is sensitive to token order even at initialisation.

4.4 Experiment 3: Prefix Length Scaling

We vary the prefix length $L \in \{8, 16, 32, 64, 128\}$ by interpolating the planner's hidden states.

Table 3: Fluency loss as a function of prefix length.

Prefix Length L	**Fluency Loss**	**Coherence Loss**
8	6.785	—
16	6.590	—
32	6.335	—
64	6.111	—
128	7.042	0.2403

Fluency improves monotonically from $L = 8$ to $L = 64$, then increases at $L = 128$ where the prefix competes with the text context for attention. We fix L to the natural planner output length in subsequent experiments.

4.5 Experiment 4: Projection Architecture Sweep

We compare four projection families trained on 30 samples with the contrastive objective (margin $m = 2.0$) for one mini-epoch of 30 steps. We report the **initial self-shuffle separation** $\bar{d} = \mathbb{E}[\|\hat{Z} - \hat{Z}_\pi\|_F^2]$ averaged over the 30 training samples. Higher $\bar{d}$ means the projection already pushes permuted plans apart; lower $\bar{d}$ means the projection is *inside the margin* and receives full contrastive gradient — i.e., more active learning signal.

Table 4: Projection architecture comparison after 30-step mini-epoch. $\bar{d} = 2.0 - \text{HingeLoss}$ (estimated from hinge loss at m=2.0). Lower $\bar{d}$ = more active gradient; higher speed = more efficient.

Architecture	**Hinge Loss**	**Est. Sep. $\bar{d}$**	**Speed (ms/step)**
Standard MLP	0.3820	1.618	64.5
Wide MLP	0.1658	1.834	72.8
Residual MLP	0.3285	1.672	81.3
Gated MLP	**1.5786**	**0.421**	75.0

The gated architecture has the lowest initial separation ($\bar{d} = 0.42$), keeping it inside the margin and receiving maximal contrastive gradient throughout training. Standard, wide, and residual projections already separate beyond the margin at initialisation, giving them near-zero gradient from step one. We therefore use the gated projection for Experiment 5b, where learning signal matters most.

4.6 Experiment 5: Representational Geometry Mapping

To understand how semantic utility and structural sensitivity interact, we employ a phased training protocol that maps the projection's representational geometry across three distinct optimization

regimes.

Regime 1: Pure Alignment. We first train the projection exclusively on an alignment objective: minimizing the MSE between the projected latent plan $\hat{Z}$ and the decoder's own token embeddings for the target text. This teaches the projection to speak the decoder's language.

Table 5: Regime 1: Semantic utility established after 1200 steps of pure alignment.

Condition	**Fluency Loss**	**Δ vs. Real**
Real $\hat{Z}$	4.689	—
Zeroed $\hat{Z}$	6.298	$\Delta_z = +1.609$

The alignment phase successfully establishes strong semantic compatibility ($\Delta_z = +1.61$ nats). The decoder actively prefers the projected plan over silence, proving that the latent projection manifold can encode useful semantic constraints.

Regime 2: Full Contrastive Optimization. We then apply our order-aware contrastive loss (Equation 3, $m = 4.0$), allowing the optimizer to update all projection parameters to maximize structural separation between ordered and shuffled plans.

Table 6: Regime 2: Full contrastive optimization induces structure sensitivity but destroys alignment.

Condition	**Fluency Loss**	**Δ vs. Real**
Real $\hat{Z}$	7.176	—
Zeroed $\hat{Z}$	5.785	$\Delta_z = -1.391$
Shuffled $\hat{Z}$	7.159	$\Delta_s = -0.017$
Wrong-structure penalty $S = -0.017 - (-1.391) = +\mathbf{1.374}$		

Full contrastive optimization successfully induces deep structural sensitivity ($S = +1.374$ nats). However, it triggers catastrophic semantic drift: Δ_z collapses from $+1.61$ back to -1.39. The optimizer escapes the decoder's embedding manifold entirely to maximize shuffle separability, creating structure that is distinct but semantically meaningless to the decoder.

Regime 3: Orthogonal Dual-Head Architecture. To prevent semantic collapse, we design an orthogonal dual-head architecture: $Z_{fused} = Z_{sem} + \alpha Z_{struct}$. We freeze the aligned semantic head Z_{sem} and train only the structural head Z_{struct} with an explicit orthogonality penalty ($\cos(Z_{sem}, Z_{struct})^2 \to 0$).

This regime completely resolves the catastrophic drift, preserving the semantic alignment ($\Delta_z = +1.434$). However, the structural penalty vanishes ($S < 0$). This failure to separate ($S = -1.563$) is a fundamental property of the architectural constraint rather than simply insufficient training: because the structural head is restricted to point-wise, additive perturbations that must remain orthogonal to the fixed semantic vectors, it lacks the topological freedom required to globally rearrange the manifold to encode sequence-level order sensitivity.

Table 7: Regime 3: Orthogonal dual-head preserves alignment but lacks structural capacity.

Condition	**Fluency Loss**	**Δ vs. Real**
Real $\hat{Z}$	4.351	—
Zeroed $\hat{Z}$	5.785	$\Delta_z = +\mathbf{1.434}$
Shuffled $\hat{Z}$	4.221	$\Delta_s = -0.129$
Wrong-structure penalty $S = -0.129 - (+1.434) = -1.563$		

5 Discussion

The Semantic-Structural Tension. Our experiments reveal a fundamental architectural tradeoff in explicit latent planning: **semantic compatibility and structural discrimination occupy competing optimization regimes in the projection manifold.** When the projection is forced to encode global ordering, it deforms the continuous space so severely that it loses alignment with the decoder's discrete token embeddings. When the space is geometrically locked to preserve alignment (via gating or orthogonality), the projection lacks the capacity to induce deep structural sensitivity.

The NaN Problem. Mixed-precision soft-prompting on MPS exposes a subtle interaction: the frozen fp16 GPT-2-XL SDPA kernel in `train()` mode produces NaN activations through a double-gradient accumulation path via the LayerNorm following the projection. Our protocol (Section 3.3) resolves this entirely for all architectural variants, enabling reliable gradient flow through fp32 embedding-space contrastive objectives.

Limitations & Future Work. Our experiments deliberately employ a constrained scale (up to 450 training pairs) to map projection geometries without extensive compute. While sufficient for falsifying the hypothesis that simple gating or additive orthogonality can solve the semantic-structural tension, we cannot rule out that the semantic-structural tension is partially attributable to underfitting at this scale, and scaling experiments remain necessary to confirm the finding is architectural rather than data-regime dependent. Future work must address the core question identified here: how to encode order-sensitive latent trajectories without destroying decoder-compatible semantic geometry.

6 Related Work

Diffusion + Autoregressive Hybrids. Lovelace et al. [8] propose Latent Diffusion for Language, which encodes sentences into a continuous latent space and decodes with an autoregressive LM. Our work differs in using the hidden states of a diffusion-based masked LM directly as a soft conditioning prefix rather than sampling from the latent space.

Plan-then-Generate. Yao et al. [9] and Wei et al. [10] demonstrate that multi-step reasoning improves generation quality. Genesis Directive externalises the planning stage entirely into a separate model, allowing independent scaling of the planner.

Prefix Tuning. Li and Liang [3] and Lester et al. [11] parametrise prefix vectors as learnable embeddings. We treat the prefix as dynamically computed from a prompt-conditioned planner, making it task-adaptive without per-task fine-tuning.

7 Conclusion

We presented **Genesis Directive**, a framework for mapping the representation geometry of explicit latent planning channels in autoregressive text generation. Our experiments isolating semantic alignment from order-aware contrastive optimization reveal a fundamental representational tradeoff:

- The projection manifold can easily achieve strong semantic compatibility ($\Delta_z = +1.61$ nats), proving the decoder can actively utilize the latent plan.
- The projection can also learn deep structural sensitivity ($S = +1.37$ nats) by penalizing scrambled token order.
- However, **these objectives compete for the same latent coordinates**. Full contrastive optimization induces catastrophic semantic drift, while structurally constrained orthogonal dual-head architectures preserve semantics but lack the capacity for deep structural separation.
- A systematic protocol for mixed-precision training on Apple MPS eliminates fp16 NaN gradients entirely, providing a broadly applicable engineering contribution for researchers working with frozen fp16 LLMs.

Genesis Directive establishes that simple gating or additive orthogonality is insufficient to solve the semantic-structural tension in latent-prefix coupling. Future work must focus on deeper architectural unbinding to encode order-sensitive latent trajectories without destroying decoder-compatible semantic geometry.

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